SIMATS ENGINEERING
LAB EXPERIMENTS

COURSE NAME: DATABASE MANAGEMENT SYSTEMS
COURSE CODE: CSA0515

EXP 2: DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK, DEFAULT, NOT NULL.

```
mysql> CREATE DATABASE ShopDB;
Query OK, 1 row affected (0.06 sec)

mysql> USE ShopDB;
Database changed
mysql> CREATE TABLE Category (Cat_ID INT PRIMARY KEY, Cat_Name VARCHAR(40) NOT NULL UNIQUE);
Query OK, 0 rows affected (0.09 sec)

mysql> DESC Category;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| Cat_ID   | int         | NO   | PRI | NULL    |       |
| Cat_Name | varchar(40) | NO   | UNI | NULL    |       |
+----------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)

mysql> CREATE TABLE Products (Prod_ID INT PRIMARY KEY, Prod_Name VARCHAR(50) NOT NULL, Price DECIMAL
(8,2) CHECK (Price > 0), Stock INT DEFAULT 10, Barcode VARCHAR(20) UNIQUE, Cat_ID INT, FOREIGN KEY (
Cat_ID) REFERENCES Category(Cat_ID));
Query OK, 0 rows affected (0.10 sec)

mysql> DESC Products;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| Prod_ID   | int          | NO   | PRI | NULL    |       |
| Prod_Name | varchar(50)  | NO   |     | NULL    |       |
| Price     | decimal(8,2) | YES  |     | NULL    |       |
| Stock     | int          | YES  |     | 10      |       |
| Barcode   | varchar(20)  | YES  | UNI | NULL    |       |
| Cat_ID    | int          | YES  | MUL | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
6 rows in set (0.05 sec)
```

```
mysql> DESC Products;INSERT INTO Category VALUES (1, 'Electronics'), (2, 'Furniture'), (3, 'Statione
ry');
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| Prod_ID   | int          | NO   | PRI | NULL    |       |
| Prod_Name | varchar(50)  | NO   |     | NULL    |       |
| Price     | decimal(8,2) | YES  |     | NULL    |       |
| Stock     | int          | YES  |     | 10      |       |
| Barcode   | varchar(20)  | YES  | UNI | NULL    |       |
| Cat_ID    | int          | YES  | MUL | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> INSERT INTO Products VALUES (101, 'Keyboard', 750.00, 20, 'BC101', 1), (102, 'Office Chair',
3500.00, 5, 'BC102', 2), (103, 'Notebook', 80.00, DEFAULT, 'BC103', 3);
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Category;
+--------+-------------+
| Cat_ID | Cat_Name    |
+--------+-------------+
|      1 | Electronics |
|      2 | Furniture   |
|      3 | Stationery  |
+--------+-------------+
3 rows in set (0.00 sec)

mysql> SELECT * FROM Products;
+---------+--------------+---------+-------+---------+--------+
| Prod_ID | Prod_Name    | Price   | Stock | Barcode | Cat_ID |
+---------+--------------+---------+-------+---------+--------+
|     101 | Keyboard     |  750.00 |    20 | BC101   |      1 |
|     102 | Office Chair | 3500.00 |     5 | BC102   |      2 |
|     103 | Notebook     |   80.00 |    10 | BC103   |      3 |
+---------+--------------+---------+-------+---------+--------+
3 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Products VALUES (101, 'Mouse', 500.00, 10, 'BC104', 1);
ERROR 1062 (23000): Duplicate entry '101' for key 'products.PRIMARY'
mysql> INSERT INTO Products VALUES (104, 'Speaker', 1200.00, 10, 'BC101', 1);
ERROR 1062 (23000): Duplicate entry 'BC101' for key 'products.Barcode'
mysql> INSERT INTO Products VALUES (105, 'Monitor', -500.00, 5, 'BC105', 1);
ERROR 3819 (HY000): Check constraint 'products_chk_1' is violated.
mysql> INSERT INTO Products VALUES (106, 'Lamp', 400.00, 10, 'BC106', 99);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`shopdb`.`prod
ucts`, CONSTRAINT `products_ibfk_1` FOREIGN KEY (`Cat_ID`) REFERENCES `category` (`Cat_ID`))
mysql> INSERT INTO Products VALUES (107, NULL, 200.00, 10, 'BC107', 1);
ERROR 1048 (23000): Column 'Prod_Name' cannot be null
mysql> INSERT INTO Products (Prod_ID, Prod_Name, Price, Barcode, Cat_ID) VALUES (108, 'USB Cable', 2
50.00, 'BC108', 1);
Query OK, 1 row affected (0.05 sec)

mysql> SELECT * FROM Products WHERE Prod_ID = 108;
+---------+-----------+--------+-------+---------+--------+
| Prod_ID | Prod_Name | Price  | Stock | Barcode | Cat_ID |
+---------+-----------+--------+-------+---------+--------+
|     108 | USB Cable | 250.00 |    10 | BC108   |      1 |
+---------+-----------+--------+-------+---------+--------+
1 row in set (0.00 sec)
```